



IXL Skill Alignment

Accelerated 7th grade alignment for Math Nation: Florida's B.E.S.T. Core Math Curriculum



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Unit 1

Equations and Inequalities

Textbook section	IXL skills
Lesson 1.1: Rewriting Numbers in Equivalent Forms	1. Convert fractions or mixed numbers to decimals M2D <i>Also consider</i> - Convert between mixed numbers and improper fractions LGK - Convert between percents, fractions, and decimals 2HW
Lesson 1.2: Converting Between Repeating Decimals and Fractions	1. Write a repeating decimal as a fraction W49 <i>Also consider</i> Convert decimals to fractions or mixed numbers HZD
Lesson 1.3: Solving Two-Step Equations Using Properties of Equality	1. Solve two-step equations: complete the solution X2L <i>Also consider</i> - Which x satisfies an equation? DJS - Solve two-step equations using algebra tiles R9E
Lesson 1.4: Solving Two-Step Equations	1. Solve two-step equations without parentheses CMX 2. Solve equations of the form $px + q = r$ with fractions Y6Q
Lesson 1.5: Solving Mathematical and Real-World Problems with Equations	1. Solve two-step equations with parentheses NSH 2. Solve equations of the form $p(x + q) = r$ with fractions A8D 3. Choose two-step equations: word problems 8NH <i>Also consider</i> - Solve two-step equations with integers QEB - Solve equations involving like terms VSW

Lesson 1.6: Writing Two-Step Equations from Real-World Contexts

1. Solve two-step equations: word problems D2Y

Lesson 1.7: Writing and Solving Equations in Real-World Context

Lesson 1.8: Determining Solutions to Inequalities

1. Solutions to inequalities G2U

2. Solve two-step inequalities N9D

3. Graph solutions to two-step inequalities WHT

Lesson 1.9: Solving Inequalities - Part 1

1. Solve multi-step inequalities 6AZ

2. Graph solutions to multi-step inequalities HKW

Lesson 1.10: Solving Inequalities - Part 2

3. Solve multi-step inequalities: word problems DKH

Also consider

- Write inequalities from number lines YW9
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Unit 2

Rational and Irrational Numbers

Textbook section	IXL skills
Lesson 2.1: Square Values	1. Identify square numbers 54P 2. Evaluate squares H77 3. Complete a function table: quadratic functions VN8
Lesson 2.2: Square Values and Square Roots	1. Solve equations using square roots NNA 2. Square roots of perfect squares 9RS <i>Also consider</i> • Relationship between squares and square roots 8W2
Lesson 2.3: Cube Values	1. Evaluate cubes AMR 2. Cube roots of positive and negative perfect cubes J7K 3. Solve equations using cube roots TQ5
Lesson 2.4: Approximating Irrational Numbers	1. Estimate square roots and cube roots PFD 2. Identify rational and irrational numbers NV6 <i>Also consider</i> • Square roots and cube roots 9LU • Identify rational and irrational square roots UGE
Lesson 2.5: Rational and Irrational Numbers on the Number Line	1. Irrational numbers on number lines QZ7
Lesson 2.6: Plotting, Ordering, and Comparing Real Numbers	1. Put rational numbers in order QP5 2. Compare and order rational and irrational numbers EED <i>Also consider</i> • Compare rational numbers MUK

Unit 3

Exponents and Scientific Notation

Textbook section	IXL skills
Lesson 3.1: Product Laws of Exponents	1. Evaluate powers SZQ 2. Multiplication with exponents 5SK 3. Power rule 8DA <i>Also consider</i> • Understanding exponents BFA
Lesson 3.2: Quotient Laws of Exponents	1. Powers with negative bases WFW <i>Also consider</i> • Division with exponents XNC • Identify equivalent expressions involving whole-number exponents KBE
Lesson 3.3: Exponential Expressions	1. Evaluate expressions using properties of exponents ZE9
Lesson 3.4: Negative Exponent Law	1. Understanding negative exponents YBB 2. Evaluate powers with negative exponents WGS
Lesson 3.5: Applying Exponent Laws	1. Identify equivalent expressions involving exponents I VLM
Lesson 3.6: Evaluating Numerical Expressions with Integer Exponents	1. Evaluate expressions using properties of exponents 6T7
Lesson 3.7: Equivalent Expressions	1. Identify equivalent expressions involving exponents II QDM
Lesson 3.8: Writing Large Numbers in Scientific Notation	1. Convert between standard and scientific notation: large numbers 2AL 2. Compare large numbers WEZ <i>Also consider</i> • Powers of ten LJQ

Lesson 3.9: Writing Small Numbers in Scientific Notation

1. Convert between standard and scientific notation: small numbers 85M
2. Compare small numbers KDR

Also consider

- Powers of ten with negative exponents SKD
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Unit 4

Solving Problems with Rational Numbers

Textbook section	IXL skills
Lesson 4.1: Adding and Subtracting Numbers Expressed in Scientific Notation	1. Add and subtract numbers written in scientific notation HUR
Lesson 4.2: Multiplying Numbers Expressed in Scientific Notation	1. Multiply numbers written in scientific notation YZU
Lesson 4.3: Dividing Numbers Expressed in Scientific Notation	1. Divide numbers written in scientific notation SGT
Lesson 4.4: Solving Real-World Problems Involving Numbers Expressed in Scientific Notation	1. Add, subtract, multiply, and divide numbers written in scientific notation: word problems D5S
Lesson 4.5: Evaluating Expressions with Radicals	1. Evaluate radical expressions D8J
Lesson 4.6: Evaluating Expressions Using the Order of Operations	
Lesson 4.7: Evaluating Real-World Problems Using the Order of Operations - Part 1	1. Evaluate variable expressions: word problems DPG
Lesson 4.8: Evaluating Real-World Problems Using the Order of Operations - Part 2	

Unit 5

Solving Multi-Step Problems with Proportional Relationships

Textbook section	IXL skills
Lesson 5.1: Scale Drawings	1. Scale drawings: find the scale factor, missing side, or missing angle WEA
Lesson 5.2: Scale Factors - Perimeter & Area	1. Perimeter and area: changes in scale ZC6
Lesson 5.3: Solving Problems Involving Perimeter and Area of Geometric Figures	1. Scale drawings: find the scale factor or missing side XHM
Lesson 5.4: Solving Problems Involving Real-World Scale Drawings	1. Scale drawings: word problems 84H <i>Also consider</i> • Scale drawings: scale factor word problems KCM
Lesson 5.5: Paying Your Way in Mathlandia	
Lesson 5.6: Currency Conversions	
Lesson 5.7: Converting Units of Length	1. Convert between customary and metric systems E8Z
Lesson 5.8: Converting Mass and Area	<i>Also consider</i>
Lesson 5.9: Converting Units of Volume	• Compare and convert customary units 4B8 • Compare and convert metric units HW2
Lesson 5.10: Exploring Multi-Step Proportion Problems	1. Compare rates: word problems VVS 2. Multi-step problems with customary or metric unit conversions UCT
Lesson 5.11: Solving Multi-Step Proportion Problems	

Unit 6

Relationships in Triangles

Textbook section	IXL skills
Lesson 6.1: The Triangle Inequality Theorem and the Converse of the Pythagorean Theorem Lesson 6.2: Applying the Triangle Inequality Theorem	1. Triangle inequality 2N5 2. Converse of the Pythagorean theorem: is it a right triangle? EQZ
Lesson 6.3: Pythagorean Theorem	1. Pythagorean theorem: find the missing leg or hypotenuse length MTM
Lesson 6.4: Using the Pythagorean Theorem to Solve Real-World Problems	1. Pythagorean theorem: word problems 87U
Lesson 6.5: Finding Distances on the Coordinate Plane	1. Find the distance between two points on a horizontal or vertical line L9V 2. Find the distance between two points ZBP
Lesson 6.6: Solving Real-World Problems on a Coordinate Plane with the Pythagorean Theorem	1. Pythagorean theorem: find the perimeter VGE
Lesson 6.7: Proportional Relationships and Similar Triangles	1. Side lengths of similar triangles QSK
Lesson 6.8: Solving Real-World Problems Using Similar Triangles	1. Similar triangles and indirect measurement 88W <i>Also consider</i> Identify similar triangles 6PD

Unit 7

Representing Proportional Relationships

Textbook section	IXL skills
Lesson 7.1: Graphing Proportional Relationships	1. Graph a proportional relationship: word problems FJW
Lesson 7.2: Constant of Proportionality - Part 1	1. Interpret graphs of proportional relationships RMH 2. Find the constant of proportionality from a table LKZ 3. Find the constant of proportionality from a graph ZUT
Lesson 7.3: Constant of Proportionality - Part 2	
Lesson 7.4: Determining if Relationships are Proportional	1. Identify proportional relationships by graphing AAN 2. Identify proportional relationships in tables and graphs VZK
Lesson 7.5: Representing Proportional Relationships with Equations	1. Write equations for proportional relationships from tables 6GU 2. Write equations for proportional relationships from graphs JKH <i>Also consider</i> Write and solve equations for proportional relationships VKK
Lesson 7.6: Translating Between Representations of Proportional Relationships	
Lesson 7.7: Putting It All Together	1. Compare proportional relationships represented in different ways 5A9

Unit 8

Relationships in Circles

Textbook section	IXL skills
Lesson 8.1: Exploring Relationships in Circles	1. Center, radius, and diameter of a circle 7RY
Lesson 8.2: Finding Circumference	2. Circumference of circles 9FM
Lesson 8.3: Finding Radius and Diameter	1. Find the radius or diameter of a circle given the circumference 2CM
Lesson 8.4: Exploring the Area of a Circle	
Lesson 8.5: Finding the Area of a Circle	1. Area of circles YA8
Lesson 8.6: Using Circumference and Area to Solve Problems	1. Circles: word problems P56
Lesson 8.7: Relationships in Circles	
Lesson 8.8: Sectors of a Circle	1. Use fractions to find the measure of an angle V76
Lesson 8.9: Creating and Interpreting Circle Graphs	1. Circle graphs and central angles KXS 2. Interpret circle graphs WNJ

Unit 9

Surface Area and Volume

Textbook section	IXL skills
Lesson 9.1: Finding Surface Area of Right Circular Cylinders Using Nets	1. Nets of three-dimensional figures 3R2
Lesson 9.2: Finding Surface Area of Right Circular Cylinders	1. Surface area of cylinders CYQ 2. Lateral area of cylinders C8K
Lesson 9.3: Solving Problems Involving Surface Area of Right Circular Cylinders	
Lesson 9.4: Volume of Right Circular Cylinders	1. Volume of cylinders FHC
Lesson 9.5: Problem-Solving with Volume of Cylinders	

Unit 10

Linear Relationships

Textbook section	IXL skills
Lesson 10.1: Proportional Relationships	- Find the constant of proportionality from a table ZCK - Identify proportional relationships by graphing RXD - Identify proportional relationships Z64 <i>Also consider</i> - Write equations for proportional relationships from tables and graphs VXM - Interpret graphs of proportional relationships Q96
Lesson 10.2: Exploring Slope	- Find the slope from two points ZAC <i>Also consider</i> - Find the slope from a table GSB
Lesson 10.3: Finding Slope Using Graphs and Tables	- Find the slope from a graph Q94 - Find the slope from a table GSB
Lesson 10.4: Slope in Real-World Contexts	- Rate of change of a linear function: graphs ZPF - Rate of change of a linear function: tables 8FE
Lesson 10.5: Exploring Equations of Lines in Slope-Intercept Form	- Slope-intercept form: find the slope and y-intercept U55 - Graph a line using slope FSV - Graph a line from an equation in slope-intercept form W5E <i>Also consider</i> - Write a linear equation from a slope and y-intercept WHP - Compare linear functions: tables, graphs, and equations N7D

Lesson 10.6: Graphing Lines from Tables and Real-World Contexts

1. Complete a table and graph a linear function DC2
2. Complete a table and make a graph: word problems T7R

Also consider

- Complete a table for a linear function D9B
- Graph a line from an equation in standard form 7MZ

Lesson 10.7: Interpreting Linear Relationships from Graphs and Equations

1. Interpret the slope and y-intercept of a linear function H5B

Also consider

- Interpret points on the graph of a linear function 9E8

Lesson 10.8: Interpreting Linear Relationships from Tables and Written Descriptions**Lesson 10.9:** Writing Equations of Linear Relationships

1. Write a linear equation from a graph WHM
2. Write a linear function from a table UYY

Also consider

- Write a linear equation from a slope and a point VKP
- Write a linear equation from two points 2R9

Lesson 10.10: Writing Equations for Real-World Contexts

1. Write linear functions: word problems YK6

Unit 11

Solving Equations and Systems of Equations

Textbook section	IXL skills
Lesson 11.1: Solving Equations	1. Solve equations involving like terms Q2B 2. Solve equations using the distributive property QWE 3. Solve two-step equations: word problems 8CZ
Lesson 11.2: Solving Equations with Variables on Both Sides	1. Solve equations with variables on both sides ZYL
Lesson 11.3: Solutions to Linear Equations	1. Find the number of solutions XDE <i>Also consider</i> • Create equations with no solutions or infinitely many solutions 7TY
Lesson 11.4: Solving Multi-Step Equations	1. Solve multi-step equations 55K <i>Also consider</i> • Solve multi-step equations: complete the solution PGH • Solve equations: mixed review HZZ
Lesson 11.5: Introduction to Systems of Two Linear Equations	1. Is (x, y) a solution to the system of equations? N46 2. Find the number of solutions to a system of equations by graphing AGZ <i>Also consider</i> • Solve a system of equations in slope-intercept form 9Z8
Lesson 11.6: Solutions to Systems of Equations	
Lesson 11.7: Solving Systems by Graphing	1. Solve a system of equations by graphing C8X <i>Also consider</i> • Write a system of equations given a graph TS7
Lesson 11.8: Solving Systems of Equations by Graphing for Real-World Contexts	1. Solve a system of equations by graphing: word problems W9J

Unit 12

Functions

Textbook section	IXL skills
Lesson 12.1: Exploring Relations	1. Relations: convert between mappings, tables, graphs, and ordered pairs LQJ
Lesson 12.2: Identifying Domain and Range	1. Domain and range of relations T7P
Lesson 12.3: Is It a Function?	1. Identify functions JT2 2. Identify functions: graphs AEB <i>Also consider</i> Identify independent and dependent variables FSF
Lesson 12.4: Interpreting Graphs of Functions	1. Identify graphs: word problems HWC
Lesson 12.5: Sketching Graphs of Functions	
Lesson 12.6: Analyzing Descriptions and Graphs	
Lesson 12.7: Exploring Functions Using Equations, Tables, and Graphs	1. Complete a function table from an equation ESE 2. Complete a table for a function graph 7EK <i>Also consider</i> Find values using function graphs 7N2
Lesson 12.8: Is it Linear?	1. Identify linear and nonlinear functions: graphs and equations XB8
Lesson 12.9: Using Tables to Determine if a Function is Linear	1. Identify linear and nonlinear functions: tables VGS

Unit 13

Representing Data

Textbook section	IXL skills
Lesson 13.1: Representing Real-World Contexts with Graphs	1. Choose the best type of graph VP6
Lesson 13.2: Creating Displays for Numerical Data	1. Create dot plots 22B 2. Create stem-and-leaf plots 8AP 3. Create histograms LGG <i>Also consider</i> - Box plots SKN - Create and interpret dot plots with fractions SHS
Lesson 13.3: Creating Displays for Categorical Data	1. Create bar graphs E76
Lesson 13.4: Representing Bivariate Data	
Lesson 13.5: Constructing Line Graphs	1. Create line graphs ERM <i>Also consider</i> Interpret line graphs TGQ
Lesson 13.6: Constructing Scatter Plots	1. Create scatter plots AVL 2. Describe scatter plots CG8 3. Describe associations between two variables using scatter plots 9ER
Lesson 13.7: Informally Fitting a Line to Bivariate Data	1. Outliers in scatter plots RP8 2. Identify lines of best fit BG7
Lesson 13.8: Scatter Plots and Lines of Fit	<i>Also consider</i> Make predictions with scatter plots CM7
Lesson 13.9: Interpreting Linear Equations Fit to Bivariate Data	1. Interpret lines of best fit: word problems 9DW

Also consider

- Write equations for lines of best fit ZQ6

Unit 14

Equivalent Algebraic Expressions

Textbook section	IXL skills
Lesson 14.1: Applying Exponent Laws to Algebraic Expressions - Part 1	1. Multiply powers: variable bases TR9 2. Powers of a power: variable bases 2CU
Lesson 14.2: Applying Exponent Laws to Algebraic Expressions - Part 2	1. Divide monomials with positive exponents C8L
Lesson 14.3: Algebraic Expressions with Negative Exponents	1. Variables with negative exponents 2NL
Lesson 14.4: Applying Multiple Laws of Exponents	1. Multiply and divide powers: variable bases 85P
Lesson 14.5: Applying the Distributive Property with Variables	1. Multiply using the distributive property: area models 8P3
Lesson 14.6: Multiplying Algebraic Expressions	2. Multiply linear expressions with rational coefficients 9B9
Lesson 14.7: Factoring Common Monomial Factors	1. Common monomial factors K8S
Lesson 14.8: Rewriting Expressions with Common Monomial Factors	2. Factor out a monomial BFZ

Unit 15

Transformations

Textbook section	IXL skills
Lesson 15.1: Slides, Flips, and Spins	1. Describe how a figure moves WER <i>Also consider</i> Congruence statements and corresponding parts LPP
Lesson 15.2: Reflections and Rotations	1. Identify reflections and rotations S8S 2. Rotate polygons about a point A5K
Lesson 15.3: Translations	1. Identify translations HJ9 2. Translations: graph the image XUS 3. Translations: find the coordinates RUP
Lesson 15.4: Reflections on the Coordinate Plane	1. Reflections: graph the image QLJ 2. Reflections: find the coordinates AQ8
Lesson 15.5: Rotations on the Coordinate Plane	1. Rotations: graph the image AC9 2. Rotations: find the coordinates HHS
Lesson 15.6: Describing Rigid Transformations	1. Describe transformations VAY <i>Also consider</i> Determine if two figures are congruent: justify your answer HJU
Lesson 15.7: Dilations	1. Classify dilations JTS 2. Scaled polygons 9ZA
Lesson 15.8: Dilations on the Coordinate Plane	1. Dilations: graph the image 9T4 2. Dilations: find the coordinates UV9 <i>Also consider</i> - Similar and congruent figures U85 - Determine if two figures are similar: justify your answer CWB



Lesson 15.9: Scale Factors of Dilations

1. Dilations: find the scale factor 8NK

Unit 16

Properties and Theorems of Angles

Textbook section	IXL skills
Lesson 16.1: Angle Relationships	1. Identify complementary, supplementary, vertical, adjacent, and congruent angles HGV 2. Find measures of complementary, supplementary, vertical, and adjacent angles R2B 3. Write and solve equations using angle relationships DBV
Lesson 16.2: Angle Relationships in Triangles	1. Find missing angles in triangles JFJ 2. Exterior Angle Theorem FMP 3. Triangle Angle-Sum Theorem 6Q6
Lesson 16.3: Interior Angles of a Polygon	1. Interior angles of polygons JBP
Lesson 16.4: Finding Missing Angles in Polygons	

Unit 17

Probability

Textbook section	IXL skills
Lesson 17.1: Determining Sample Space Using Lists and Tree Diagrams	1. Compound events: find the number of outcomes P5R 2. Samples spaces for repeated events AGW
Lesson 17.2: Calculating Theoretical Probability	1. Probability of repeated experiments BDZ <i>Also consider</i> Probability of simple events 5ZY
Lesson 17.3: Calculating Theoretical Probability of Compound Events	1. Experimental probability FRL
Lesson 17.4: Applying Proportional Reasoning to Probability	1. Make predictions 38C 2. Write a discrete probability distribution ELU
Lesson 17.5: Solving Problems Using Probability	
Lesson 17.6: Making Predictions	